A Novel Distributed Situation Awareness Consensus Approach for UAV Swarm Systems

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***Abstract*— This paper develops a novel approach to addressing the distributed situation awareness (SA) consensus problem for unmanned aerial vehicle (UAV) swarm systems. SA consensus is an important condition for gaining decision-making superiority. However, because of the complexity and antagonism of the mission environment, the widely employed centralized architectures are unable to reach a distributed consensus. We address this urgent issue by proposing a systematic distributed SA consensus scheme, including a distributed optimization-based consensus reaching model, dual-loop decision-making framework, and novel coordination algorithm. Necessarily, we conduct convergence analysis on the proposed algorithm. The effectiveness and superiority of the proposed method are verified by comparative simulations.**

***Index Terms*— Consensus, distributed optimization, decision-making, situation awareness, unmanned aerial vehicle (UAV) swarm systems.**

I. Introduction

HIGH mobility and integration of advanced technologies have accelerated a wide range of applications of unmanned aerial vehicle (UAV) in urban areas, such as traffic monitoring, infrastructure inspection, and target tracking [1], [2], [3], [4]. As one of the most emergent topics derived from ergonomics, situation awareness (SA) is initially concerned with one's perception of their surroundings [5]. For intelligent UAVs that extricate humans from direct involvement in risky missions, SA represents the knowledge about surroundings that enable appropriate decision-making [6].

Indeed, a swarm of UAVs could provide an accurate multi-perspective view of a real-world scenario [7]. However, each vehicle within a group always holds its own SA during the cooperative operation processes, i.e., variation in individual cognition will be generated even for the same target. This variation is caused by uncertainties such as flight height differences, observation angle limitations, building occlusions, and extreme weather [8]; on the other hand, it is due to system incompatibility in terms of heterogeneous sensor coexistence and multisource information fusion [9]. Considering that the inconsistency in UAVs' SA may lead to low autonomy, poor coordination, and even pose a threat to swarm system performance and security [10], a common shared vision is urgently needed, which allows multiple UAVs to have similar awareness of the same scene in the cognitive field.

Generally, the aforementioned study can be addressed as an SA consensus problem [11], where agreement can be reached among decision-makers observing the same phenomenon on the discordant opinions. Although an ideal consensus degree can enhance the acceptability of the final decision, the complexity and antagonism of the mission environment render traditional centralized architectures no longer applicable. This result shifts SA consensus from an individual's mind to a collective cognitive process. Using a system view in teamwork, distributed situation awareness (DSA) is concerned with how knowledge is used and parsed among agents in systems interaction, implying different, but potentially compatible, requirements and purposes [12]. In DSA, each agent in the system, receiving input from its sensors or communicating with others, has local awareness of a situation, which contributes to high-level knowledge structures [13]. From this perspective, there is system-level awareness of a situation such that the loss of SA by one agent can be compensated by the activity of other agents. Therefore, DSA guarantees individual autonomy moving to swarm intelligence.

In this article, we focus on the DSA consensus problem for UAV swarm systems and develop a fully distributed coordination algorithm to reach consensus with both theoretical analysis and case verification.

A. Related Works

As the mathematical model is the premise for solving a consensus reaching problem, numerous consensus models have been built in the context of heterogeneous group decision-making to support the agreement process among divergent opinions [14], [15], [16]. Apparently, there has been minimal research directly oriented to SA consensus for UAV swarm systems. Among the countable literature, D'Aniello et al. [11] integrated a multiagent fuzzy consensus model into an SA framework with multiple levels of perceptions, where a fuzzy preference relation was used to represent the opinions of decision-makers; Cavaliere et al. [17] utilized fuzzy ontology to model event aggregation, which allows UAVs to express preferences on the situations. Nevertheless, centralized architectures in the aforementioned work have

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limited scalability and are less robust to local uncertainties, which is unapplicable to large-scale UAV swarm systems with a high degree of autonomy. Gao et al. [18] investigated a distributed SA model for a UAV swarm system and further developed a cooperative SA consensus model in [19], in which UAVs were regarded as decision-makers with a certain autonomous intelligence. By considering the uncertainty of information, Zhu et al. [20] established an SA consensus model for a swarm of cooperative UAVs, in which the consensus at different levels of perception, comprehension, and prediction were analyzed. Although consensus can be achieved in the above models, fully distributed interactions among decision-makers were not implemented. Consensus was obtained based on regulating the consensus degree in a centralized manner rather than with local information exchanges. In [21], the satellite-assisted space situational awareness consensus was formulated as a nonlinear consensus tracking problem, where a distributed framework was designed to update the tracking estimates and share local information. This level of consensus ensures optimal proximity of individual cognition to others, but does not guarantee agreement with the real situation.

As a longstanding topic in the optimization community, distributed optimization requires multiple agents to agree on a state that minimizes a global objective function, which is a sum of the individual objectives in a distributed manner by local computation and communication [22]. In recent decades, consensus-based distributed optimization protocols have received considerable attention in terms of theoretical analysis and technical notes in the areas of multiagent systems [23], [24], [25], [26]. By leveraging the encouraging result of applying DSA to multiagent networks [27], there is great promise in addressing DSA consensus issues by formulating a distributed optimization problem.

To address distributed optimization problems, the most commonly employed distributed gradient descent (DGD) method [28] has drawn the greatest attention due to its simplicity and ease of implementation. However, the convergence is rather slow due to the fixed step sizes. Although modifications by considering accelerated strategies [29], [30], quantification [31], and sparsification [32] have made the improved versions well suited for more realistic scenarios, existing approaches still require each agent to upload the latest locally gradient at every iteration based on the current state of the system, which is rather expensive in communication. In addition, given the applications for UAV swarm systems, as previously stated, individual recognition is potentially disrupted by both practical situations and adversarial behavior. Undoubtedly, the current state with incomplete or even inaccurate information will reduce the quality of consensus. Some resilient algorithms have been designed to refine existing distributed protocols to provide guarantees against disturbances [33], [34], [35], [36], while these approaches have not been validated in UAV swarm systems in terms of DSA consensus.

B. Motivation

On examining the existing work, we discovered that the research of SA consensus directly oriented to UAV swarm systems is still inadequate compared to its urgency in practical applications.

To close the above research gaps, the main challenges may result from three aspects.

1) The essential and fundamental work lies in an appropriate model to reach two levels of DSA consensus. The modeling process in terms of problem formulation and parameter definitions remains an open issue.
2) A decision-making framework that bridges the two levels of consensus and facilitates consensus reaching in a distributed manner is another challenging and necessary task. Such a framework not only contributes to the generation of high-quality solutions, but also resist possible disruptions.
3) A distributed coordination algorithm is an indispensable tool to obtain feasible solutions based on the consensus reaching model. Better performance is required compared to traditional methods in terms of communication efficiency.

To address these challenges, it is consequential to develop a novel systematic methodology by taking account of the above aspects. Based on our previous work [37], extracting the intrinsic properties and simpler mechanisms that contribute to group coherence in collective behavior can provide valuable insight into the establishment of complicated models. In addition, some promising results widely observed in recent years are also inspiring and attractive to us for developing more efficient operators in the design of both the decision-making framework and coordination algorithms [38], [39], [40], [41], [42].

C. Main Contributions

In this paper, a systematic scheme is proposed to achieve DSA consensus for UAV swarm systems. As a generic methodology, this proposed scheme consists of a consensus reaching model, decision-making framework, and coordination algorithm.

The main contributions are summarized as follows:

1) Following a distributed optimization formulation, A DSA consensus reaching model, in which relative consensus among decision-makers and absolute consensus between the system and the real situation are both considered, is mathematically established.
2) A dual-loop decision-making framework is proposed to support the acquisition of DSA consensus, where the feedback-based consensus weight correction mechanism between inner loop and outer loop provides guarantees against disturbances.
3) A novel coordination algorithm with a dual-mode switching operator under the feedback mechanism is developed to avoid real-time consecutive communication between UAVs with different intellectual levels.

D. Organization

The remainder of the paper is organized as follows: Section II provides the modeling of DSA consensus. Section III introduces the proposed decision-making framework to reach consensus. Section IV details the novel distributed coordination algorithm. In Section V, we present the simulation results, and conclusions are summarized in Section VI. For reference, we summarize the notations and their meanings in TABLE I.

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TABLE I
NOTATIONS AND MEANINGS

Notations	Meaning
$\boldsymbol{V}, \boldsymbol{X}$	The sets of UAVs and targets information
$\boldsymbol{P}, p$	Fuzzy preference relation matrix and its element
$\boldsymbol{P}^*, p^*$	Real matrix and its element
$\boldsymbol{S}, s$	Similarity matrix and its element
$\boldsymbol{C}, c$	Consensus matrix and its element
N, M	Number of UAVs and targets
v, x	UAVs and SA information
$k, l; i, j$	Index of UAVs and targets
μ	Membership function
t_1, t_2	Time in real-world scenario and step size of the algorithm
$\mathcal{G}; \mathcal{V}, \mathcal{E}, \mathcal{N}$	Graph and the sets of nodes, edges, and neighbors
B	Intercommunication interval
f	Individual objective function
$\hat{p}$, $\bar{p}$	Estimates of the real situation, the optimal solution
$\boldsymbol{\omega}, \boldsymbol{\omega}_l$	Consensus weight and neighborhood consensus weight
β	Adjustment factor of estimation
L	The gradient boundary
$\boldsymbol{u}$	The control input
K_P, K_I, K_D	Control gains of PID
$\boldsymbol{c}_{\mathrm{L}}^k$, $\boldsymbol{c}_{\mathrm{L}}^{\min}$, $\boldsymbol{c}_{\mathrm{L}}^{\mathrm{S}}$	Local consensus parameter, threshold, and the mode switching threshold
$\boldsymbol{M}_I$	The interaction mode criteria
n_d	The degree of the node
K_F	Objective optimization control gain
p_r	Acceptance probability
$\boldsymbol{I}$	Intellectual level
r_1, r_2	Uniformly distributed random parameters
F_{b}	The break identifier
$\boldsymbol{y}$	The vector of individual
a	The alignment weight
α	The step-size
η	The lower bound of the nonzero alignment weight
$T_{\max}^1$ / $T_{\max}^2$	Maximum simulation time of the outer/inner loop
A	A function related to η and B
E_{r}	The error between the individual average estimated global objective function value and the global objective optimal value
F, F^*	The swarm objective optimization function and the real swarm objective optimization function
c_{G}	The global consensus parameter
b	The profit of SA information
π	The normalized profit of SA information

II. MODELING OF DISTRIBUTED SITUATION AWARENESS CONSENSUS FOR UAV SWARM SYSTEMS

In this section, we describe the mission scenario. Then, the SA-oriented system model and related definitions are provided. The optimization formulation is given to mathematically describe the core issue of this paper.

A. Scenario Description

As shown in Fig.1, a swarm of N homogeneous UAVs in $\boldsymbol{V} = \{v_1, v_2, \ldots, v_N\}$ are required to implement surveillance in a given urban area [3] with M different targets, denoted as $\boldsymbol{X} = \{x_1, x_2, \ldots, x_M\}$. For each target i, $i = 1, 2, \ldots, M$, x_i represents its initial SA information, including attribute (e.g., vehicles and building), state (e.g., moving, stationary, and orientation, etc.), importance level (e.g., urgency of emergencies), and environmental data. This step is achieved based on a three-level model developed in [5].

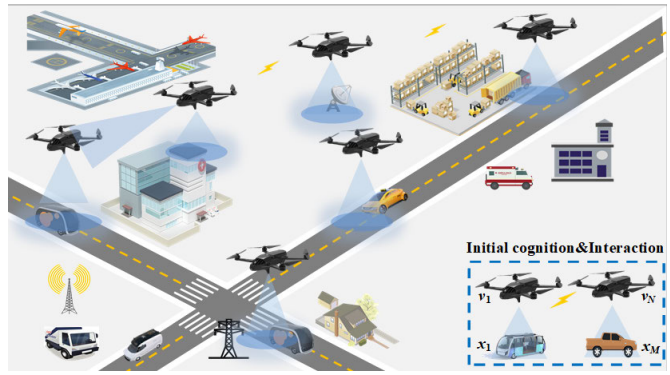

Fig. 1. Architecture of the UAV swarm system in surveillance mission implementation.

B. System Model

The fuzzy preference relation [43] is adopted to describe the UAV's preferences on a set of targets. Specifically, each $v_k, k = 1, 2, \ldots, N$ in $\boldsymbol{V}$ provides a fuzzy preference information $\boldsymbol{P}^k(t_1, t_2)$ for the targets in $\boldsymbol{X}$, where t_1 and t_2 denote the actual time in a real-world scenario and the step size of the algorithm during a consensus adjustment, respectively. Note that $\boldsymbol{P}^k(t_1, t_2)$, abbreviated as $\boldsymbol{P}^k$, is represented by the matrix $\boldsymbol{P}^k = \left(p_{ij}^k\right)_{M\times M}$, in which each element $p_{ij}^k = \mu_P(x_i, x_j)$ is interpreted as the preference degree of v_k for x_i over x_j, where the value of the membership function μ_P is within [0, 1].

Note that a common natural assumption is that $p_{ij}^k + p_{ji}^k$=1, $\forall i, j = 1, 2, \ldots, M$ [44]. Based on preference relation theory [45], the following definitions can be further given:

Definition 1: : $p_{ij}^k > 0.5$ indicates that v_k is more inclined to x_i. The closer the value of p_{ij}^k is to 1, the stronger this tendency. Conversely, $p_{ij}^k < 0.5$ implies that x_j is preferred to x_i. The closer the value is to 0, the stronger this tendency.

Definition 2: : $p_{ij}^k = 0.5$ indicates that v_k has the same tendency toward x_i and x_j.

For distributed interactions, an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is utilized to represent the topology of the information exchange among UAVs, where $\mathcal{V} = \{1, 2, \ldots, N\}$ denotes the set of nodes and $\mathcal{E} \subset \mathcal{V} \times \mathcal{V}$ denotes the set of edges. The neighbors of node $k, k = 1, 2, \ldots, N$ are defined as $\mathcal{N}_k = \{l \in \mathcal{V} : (k, l) \in \mathcal{E}\}$. Assume that two UAVs can interact if and only if they are neighbors. The above communication topology is time-varying $\mathcal{G}_t = (\mathcal{V}, \mathcal{E}_t)$, where the edge set changes over time t due to the relative motion of the UAV and the unexpected loss of communication links. In addition, we assume that $\mathcal{G}_t$ has jointly connected switching topologies and that there are no isolated nodes in the graph. According to [28], the following condition is satisfied:

Condition 1: (Bounded Intercommunication Interval): We assume that $\mathcal{E}^* = \{(k, l)k, l \in \mathcal{V}\}$ satisfies that there is $l \in \mathcal{N}_k$ infinitely many times and that the graph $(\mathcal{V}, \mathcal{E}^*)$ is connected. For all $(k, l) \in \mathcal{E}^*$ and $t > 0$, $(k, l) \in \mathcal{E}_t \cup \mathcal{E}_{t+1} \cup \ldots \cup \mathcal{E}_{t+B-1}$, where $B \geq 1$ is the intercommunication interval.

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C. Optimization-Based Problem Formulation

Due to unacceptable disagreements in the UAV's preference information, it is necessary to coordinate them by dynamically adjusting $\boldsymbol{P}^k$ to reach a consensus. In our work, relative consensus and absolute consensus are successively considered.

1) *Relative Consensus*: It is a priority to ensure that the cognition of the individual is the same as that of its neighbor, which is expressed as

$$p_{ij}^k = p_{ij}^l, \quad \forall l \in \mathcal{N}_k, \ i, j = 1, 2, \ldots, M. \tag{1}$$

1) *Absolute Consensus*: Afterward, aligning the individual's cognition with reality, which can be reflected by defining the individual objective function $f_k(\hat{p}_{ij}^k, p_{ij}^k)$ (calculations are provided in the following section), where $\hat{p}_{ij}^k$ denotes the estimates of the real situation of v_k. Each UAV in the swarm attempts to solve a specific distributed consensus optimization problem that can be formulated as

$$\bar{p}_{ij}^k(t_1) = \arg\min_{p_{ij}^k} f_k(\hat{p}_{ij}^k, p_{ij}^k) \quad \text{s.t. } p_{ij}^k = p_{ij}^l, \ \forall l \in \mathcal{N}_k, x_j \in \boldsymbol{X}^k, x_j \in \boldsymbol{X}^l \tag{2}$$

where $\bar{p}_{ij}^k(t_1)$ represents the optimal solution and the individual cognition p_{ij}^k is the quantity to be optimized. The process of minimizing $f_k(\hat{p}_{ij}^k, p_{ij}^k)$ improves the absolute consensus between individual cognition and the real situation. The restriction of $p_{ij}^k = p_{ij}^l$ makes the individual cognition p_{ij}^k and the neighbor cognition p_{ij}^l have high relative consensus.

III. Decision-Making Framework to Reach Consensus

This section explicates the proposed decision-making framework to reach DSA consensus. First, the structure of the dual-loop framework is described. Second, the two phases in the inner loop are detailed. Finally, the design method of the outer loop is given.

A. Dual-Loop Decision-Making Framework

To support the acquisition of DSA consensus in a UAV swarm system, we propose a dual-loop decision-making framework inspired from a dual-loop control method [38], a multiphase design technique [39], and feedback mechanisms [40], [41], [42]. As illustrated in Fig. 2, the consensus reaching process is divided into the following phases:

1) *Individual Cognition*: Each $v_k, k = 1, 2, \ldots, N$ perceives the target elements $i, i = 1, 2, \ldots, M$ in real time and forms an individual SA x_i. By utilizing $\boldsymbol{P}^k = \left(p_{ij}^k\right)$, each $v_k, k = 1, 2, \ldots, N$ obtains an individual cognition of situation $\boldsymbol{X}$.
2) *System Collaboration*: Individual cognition sharing is implemented through $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ in a distributed manner, where each node in $\mathcal{V}$ interacts with its neighbor relying on local information only.
3) *Outer Loop Procedure*: A feedback controller consisting of a classical PID, as shown in (7), is introduced to calibrate the trust in neighbors, the neighborhood consensus weight, denote as $\boldsymbol{w}_l, l = 1, 2, \ldots, k$, is utilized to provide target-related estimation information to the inner loop through (8).

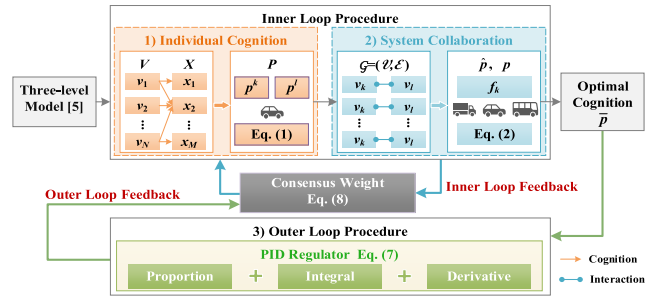

Fig. 2. Decision-making framework to reach consensus.

The consensus weight $\boldsymbol{w}$, as a bridge, coordinates individual cognition by using shared cognition as inner loop feedback, while in the outer loop, it is modified by the consistent cognition formed in the past. Thus, the input of the outer loop is the evaluation result related to the initial cognition and real situation rather than the final individual cognition. On this basis, the absolute consensus of SA is improved.

B. Individual Cognition Phase

As previously stated, it is not possible to observe all targets for each v_k in $\boldsymbol{V}$. Thus, in the first phase of the inner loop, if x_i or x_j is not observed by v_k, i.e., $x_i \notin \boldsymbol{X}^k$ or $x_j \notin \boldsymbol{X}^k$, where $\boldsymbol{X}^k \subseteq \boldsymbol{X}$ is the target set observed by v_k, p_{ij}^k does not exist, which is denoted as **Null**. If x_i and x_j can be observed by v_k, i.e., $x_i \in \boldsymbol{X}^k$ and $x_j \in \boldsymbol{X}^k$.

Based on the definitions in Section II as well as additive transitivity conditions [46], three properties are further inferred as:

Property 1: $\boldsymbol{P}^k$ has the unity of tendency, i.e., $p_{ij}^k + p_{jj'}^k = 0.5 + p_{ij'}^k$, where $x_i, x_j, x_{j'} \in \boldsymbol{X}^k$.

Property 2: The nonempty elements on $\boldsymbol{P}^k$'s main diagonal are 0.5, i.e., $p_{ii}^k = 0.5$, where $x_i \in \boldsymbol{X}^k$.

Property 3: Using the main diagonal of $\boldsymbol{P}^k$ as the axis, the sum of the two symmetrical nonempty elements is 1, i.e., $p_{ij}^k + p_{ji}^k = 1$, where $x_i, x_j \in \boldsymbol{X}^k$.

Note that Properties 2 and 3 are exceptional cases of Property 1. Specifically, when $i = j = j'$ is satisfied, Property 2 can be obtained from Property 1; when $j' = i$, Property 3 can be obtained from Property 1. Related proof is given in [47].

C. System Collaboration Phase

In the second phase, it is assumed that x_1 is the reference target known to all UAVs. According to the above properties, the other nonempty element in $\boldsymbol{P}^k$ is constructed by the first-row elements as follows:

$$p_{ij}^k = 0.5 + p_{1j}^k - p_{1i}^k \tag{3}$$

where $x_i, x_j \in \boldsymbol{X}^k$. The first item in the first row is fixed to 0.5. Therefore, we only need to adjust non-empty elements of $p_{12}^k, \ldots, p_{1(M-1)}^k$ and p_{1M}^k at the system collaboration stage.

As previously mentioned, individuals only generate estimates of the real situation related to $\boldsymbol{w}_l, l = 1, 2, \ldots, k$ and

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initial cognition p_{1j}^{l}, calculated as

$$\hat{p}_{1j}^{k}(t_1) = \begin{cases} p_{1j}^{k}(t_1, 0), \text{if} \boldsymbol{w}_l(t_1) \leq \boldsymbol{w}_k(t_1), \forall l \in \mathcal{N}_k(t_1, 0) \\ \dfrac{\sum\limits_{x_j \in \boldsymbol{X}^l \& l \in \mathcal{N}_k(t_1,0) \& \boldsymbol{w}_l(t_1) > \boldsymbol{w}_k(t_1)} p_{1j}^{l}(t_1, 0)\,(\mathbf{w}_l(t_1))^{\beta}}{\sum\limits_{x_j \in \boldsymbol{X}^l \& l \in \mathcal{N}_k(t_1,0) \& \boldsymbol{w}_l(t_1) > \boldsymbol{w}_k(t_1)} (\mathbf{w}_l(t_1))^{\beta}}, \end{cases} \tag{4}$$

where $x_j \in \boldsymbol{X}^k$, and $\beta > 0$ denotes the adjustment factor of cognition estimation. For convenience, $\hat{p}_{1j}^{k}(t_1)$ and $\boldsymbol{w}_l(t_1)$ are abbreviated as $\hat{p}_{1j}^{k}$ and $\boldsymbol{w}_l$, respectively.

Accordingly, the individual objective optimization function in (2) is denoted as $f_k(\hat{p}_{1j}^{k}, p_{1j}^{k})$ and obtained by:

$$f_k(\hat{p}_{1j}^{k}, p_{1j}^{k}) \begin{cases} = L(a_1 - p_{1j}^{k}) + (a_1)^2/2 \\ -(\hat{p}_{1j}^{k})^2 \ln(a_1), 0 \leq p_{1j}^{k} < a_1 \\ (p_{1j}^{k})^2/2 - (\hat{p}_{1j}^{k})^2 \ln(p_{1j}^{k}), a_1 \leq p_{1j}^{k} \leq \hat{p}_{1j}^{k} \\ = \dfrac{1}{2}[\hat{p}_{1j}^{k}(1 - p_{1j}^{k})/(1 - \hat{p}_{1j}^{k})]^2 - (\hat{p}_{1j}^{k})^2 \\ \cdot \ln[\hat{p}_{1j}^{k}(1 - p_{1j}^{k})/(1 - \hat{p}_{1j}^{k})], \hat{p}_{1j}^{k} < p_{1j}^{k} < a_2 \\ = L(p_{1j}^{k} - a_2) + \dfrac{1}{2}[\hat{p}_{1j}^{k}(1 - a_2)/(1 - \hat{p}_{1j}^{k})]^2 - \\ (\hat{p}_{1j}^{k})^2 \ln[\hat{p}_{1j}^{k}(1 - a_2)/(1 - \hat{p}_{1j}^{k})], \\ a_2 \leq p_{1j}^{k} \leq 1 \end{cases} \tag{5}$$

where $x_j \in \boldsymbol{X}^k$, $a_1 = [-L + \sqrt{L^2 + 4(\hat{p}_{1j}^{k})^2}]/2$, $a_2 = 1 - \frac{(\hat{p}_{1j}^{k}-1)\sqrt{(\hat{p}_{1j}^{k}-1)^2 L^2 + 4(\hat{p}_{1j}^{k})^4} - (\hat{p}_{1j}^{k}-1)^2 L}{2(\hat{p}_{1j}^{k})^2}$, $L > 0$ is the gradient boundary, and $\ln(\cdot)$ denotes the natural logarithmic function. $f_k(\hat{p}_{1j}^{k}, p_{1j}^{k})$ is strictly convex satisfying:

1) $\nabla_{p_{1j}^{k}} f_k(\hat{p}_{1j}^{k}, a_1) = \nabla_{p_{1j}^{k}} f_k(\hat{p}_{1j}^{k}, a_2) = L$.
2) $\nabla_{p_{1j}^{k}} f_k(\hat{p}_{1j}^{k}, \hat{p}_{1j}^{k}) = 0$.

To minimize $f_k(\hat{p}_{1j}^{k}, p_{1j}^{k})$ when satisfying (4), $\boldsymbol{P}^k$ is updated by the following equation:

$$p_{1j}^{k}(t_1, t_2 + 1) = p_{1j}^{k}(t_1, t_2) + \boldsymbol{u}_{1j}^{k}(t_1, t_2) \tag{6}$$

where $x_j \in \boldsymbol{X}^k$ and $\boldsymbol{u}_{ij}^{k}(t_1, t_2)$ is the control quantity to achieve consensus. For convenience, $\boldsymbol{u}_{ij}^{k}(t_1, t_2)$ is abbreviated as $\boldsymbol{u}_{ij}^{k}$ in the following sections.

D. Outer Loop Design

The accuracy of individual's estimation $\hat{p}_{1j}^{k}$ significantly affects the absolute consensus of SA. It is known from (4) that that $\hat{p}_{1j}^{k}$ is closely related to $\boldsymbol{w}_l, l = 1, 2, \ldots, k$. Thus, to improve the agreement between individual estimates and the real situation, PID feedback is introduced to modify it, which is computed by:

$$\boldsymbol{w}_k' = K_{\mathrm{P}} \frac{1}{\sum\limits_{x_j \in \boldsymbol{X}^k} f_k} + K_{\mathrm{I}} \int_0^t \frac{1}{\sum\limits_{x_j \in \boldsymbol{X}^k} f_k} \mathrm{d}t + K_{\mathrm{D}} \mathrm{d} \frac{1}{\sum\limits_{x_j \in \boldsymbol{X}^k} f_k} / \mathrm{d}t \tag{7}$$

where K_{P}, K_{I}, and K_{D} are the control gains of the proportional, integral, and differential terms, respectively; $f_k = f_k[p_{1j}^{*}(t_1), p_{1j}^{k}(t_1, 0)]$; and $p_{1j}^{*}(t_1)$ is the element of the real fuzzy preference relation matrix $\boldsymbol{P}^{*}(t_1)$, abbreviated as p_{1j}^{*} and $\boldsymbol{P}^{*}$, respectively. Note that the parameters of PID are empirically tuned. To facilitate subsequent calculations, further normalization yields

$$\boldsymbol{w} = \boldsymbol{w}_k' / \sum_{k=1}^{N} \boldsymbol{w}_k'. \tag{8}$$

IV. Proposed Distributed Coordination Algorithm

To address DSA consensus problem, a distributed coordination algorithm is presented. The convergence property is detailed to provide theoretical support.

A. Algorithm Design

1) Obtain the Local Consensus Parameter:

To reach consensus, each v_k in $\boldsymbol{V}$ is initially required to measure the local consensus parameter $\boldsymbol{c}_{\mathrm{L}}^{k}$ of SA. The procedure is summarized as follows:

Step 1: The similarity matrix $\boldsymbol{S}^{kl}$ between v_k and v_l is calculated, where $l \in \mathcal{N}_k$ and $\boldsymbol{S}^{kl} = \left(s_{ij}^{kl}\right)$. If $x_j \in \boldsymbol{X}^k$ and $x_j \in \boldsymbol{X}^l$, $s_{ij}^{kl} = 1 - \left|p_{ij}^{k} - p_{ij}^{l}\right|$; otherwise, $s_{ij}^{kl} = 1$.

Step 2: The consensus matrix $\boldsymbol{C}^k = \left(c_{ij}^{k}\right)$ is generated, where c_{ij}^{k} is computed by:

$$c_{ij}^{k} = \sum_{l \in \mathcal{N}_k} s_{ij}^{kl} / \mathrm{card}\,(\mathcal{N}_k) \tag{9}$$

where $card(\cdot) \neq 0$ denotes a function for calculating the number of elements in a set.

Step 3: The local consensus parameter $\boldsymbol{c}_{\mathrm{L}}^{k}$ is obtained by the following equation:

$$\boldsymbol{c}_{\mathrm{L}}^{k} = \left(\sum_{i=1}^{M} \left(\sum_{j=1, j \neq i}^{M} c_{ij}^{k} + c_{ji}^{k} \right) / \left(2(M-1)\right) \right) / M. \tag{10}$$

Based on the above calculations, it is determined that $\mathbf{c}_{\mathrm{L}}^{k} \in [0, 1]$. The higher the value of $\boldsymbol{c}_{\mathrm{L}}^{k}$, the better the local consensus of SA centered on v_k. Additionally, v_k keeps the current $\boldsymbol{P}^k$ unchanged when $\boldsymbol{c}_{\mathrm{L}}^{k} \geq \boldsymbol{c}_{\mathrm{L}}^{\min}$; otherwise, it will revise $\boldsymbol{P}^k$ based on (6). In particular, when the condition $\boldsymbol{c}_{\mathrm{L}}^{k} \geq \boldsymbol{c}_{\mathrm{L}}^{\min}$ is satisfied for all UAVs in $\boldsymbol{V}$, the system completes the internal collaboration.

2) Determination of the Neighbor Set:

When $\boldsymbol{c}_{\mathrm{L}}^{k} < \boldsymbol{c}_{\mathrm{L}}^{\min}$, two interaction modes can be selected for UAVs in the inner loop: the first is the egalitarian mode with $\boldsymbol{M}_{\mathrm{I}}^{k} = 0$, meaning that $\mathbf{c}_{\mathrm{L}}^{k}$ is favorable; the second is the hierarchical mode with $\boldsymbol{M}_{\mathrm{I}}^{k} = 1$, indicating that $\mathbf{c}_{\mathrm{L}}^{k}$ is weak. The network in the former is based on $\mathcal{N}_k$, while in the latter, it is a subgraph of that in the former based on the connection between individuals and adjacent superiors. If the mode switching threshold for local consensus is given as $\boldsymbol{c}_{\mathrm{L}}^{\mathrm{s}}$ satisfying $\boldsymbol{c}_{\mathrm{L}}^{\mathrm{s}} < \boldsymbol{c}_{\mathrm{L}}^{\min}$, the relationship between the interaction mode and the local consensus is expressed as follows:

$$\boldsymbol{M}_{\mathrm{I}}^{k} = \begin{cases} 1, \ \boldsymbol{c}_{\mathrm{L}}^{k} < \boldsymbol{c}_{\mathrm{L}}^{\mathrm{s}} \\ 0, \ \boldsymbol{c}_{\mathrm{L}}^{k} \geq \boldsymbol{c}_{\mathrm{L}}^{\mathrm{s}}. \end{cases} \tag{11}$$

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For each v_k, the consensus of SA is achieved based on information exchanges with its neighbor in the two modes. Then, the corresponding control input $\boldsymbol{u}_{ij}^k$ is calculated as

$$\boldsymbol{u}_{1j}^k = \sum_{l\in\mathcal{N}_k'(t_1,t_2)\&x_j\in\boldsymbol{X}^l} \min\left\{\frac{1}{n_d^k(t_1,t_2)+1}, \frac{1}{n_d^l(t_1,t_2)+1}\right\} \cdot (p_{1j}^l - p_{1j}^k) - K_{\mathrm{F}}\nabla_{p_{1j}^k} f_k(p_{1j}^k) \tag{12}$$

where $x_j \in \boldsymbol{X}^k$, $\mathcal{N}_k'(t_1,t_2)$ is the actual neighbor set of v_k, $n_d^k(t_1,t_2)$ is the degree of v_k, and the objective optimization control gain $K_{\mathrm{F}} > 0$. For convenience, $\mathcal{N}_k'(t_1,t_2)$ and $n_d^k(t_1,t_2)$ are abbreviated as $\mathcal{N}_k'$ and n_d^k, respectively.

Notably, $\mathcal{N}_k' = \mathcal{N}_k$ when the egalitarian interaction mode is chosen by v_k, i.e., $\boldsymbol{M}_{\mathrm{I}}^k = 0$; otherwise, $\mathcal{N}_k'$ is obtained by following specific steps as:

Step 1: v_k sends $\boldsymbol{w}_k$ to its neighbors in the available neighbor set $\mathcal{N}_k$. Given the following example with the consensus weight before normalization as $\boldsymbol{w} = \begin{bmatrix} 0.13607, 0.86929, 0.57970, 0.54986, 0.14495, \\ 0.85303, 0.62206, 0.35095, 0.51325, 0.40181 \end{bmatrix}$, the available topologies are described in Fig. 3(a).

Step 2: As shown in Fig. 3(b), v_k requests an interaction connection from the upper neighbor whose consensus weight is closest to its own, where the direction of the arrow indicates the direction of the interaction requests. If v_l satisfies $l \in \mathcal{N}_k$ and $\boldsymbol{w}_l \geq \boldsymbol{w}_k$, it can be recognized as the upper neighbor of v_k, i.e., it has a higher intellectual level than v_k in $\mathcal{N}_k$.

Step 3: We assume that there exists a probability for each v_k to accept the neighbor's interaction request, denoted as p_{r}. Using p_{r}=0.8 as an example, the response results of the swarm are shown in Fig. 3(c).

The actual neighbors of each UAV are shown in Fig. 3(d), where the arrow indicates the information transmission.

3) Implementation of the Algorithm:

To explain the proposed distributed coordination algorithm, the specific implementation steps are presented as follows:

Step 1: The outer loop simulation time $t_1 = 0$. $\boldsymbol{P}^*$ is generated. The initial consensus weight $\boldsymbol{w}_k(0) = \frac{1}{N}$.

Step 2: The inner loop simulation time $t_2 = 0$. The initial matrix $\boldsymbol{P}^k(t_1,0)$ of v_k is generate by the following equation:

$$p_{ij}^k(t_1,0) = \begin{cases} p_{ij}^* + r_1(1-\boldsymbol{I}_k)(1-p_{ij}^*), & \\ if\, i=1, j=2,\ldots,M, & r_2 \leq 0.5 \\ p_{ij}^* - r_1(1-\boldsymbol{I}_k)p_{ij}^*, & \\ if\, i=1, j=2,\ldots,M, & r_2 > 0.5 \\ 0.5 + p_{1j}^k(t_1,0) - p_{1i}^k(t_1,0), & otherwise \end{cases} \tag{13}$$

where $x_i, x_j \in \boldsymbol{X}^k$, $\boldsymbol{I}_k \in (0,1)$ is the intellectual level, and both $r_1 \in (0,1)$ and $r_2 \in (0,1)$ are uniformly distributed random numbers. The break identifier $F_{\mathrm{b}} = 0$. The UAV index $k = 1$. The vector of individual k at time t is defined as $\boldsymbol{y}^k(t) = \left[p_{12}^k(t_1,t_2), p_{13}^k(t_1,t_2), \ldots, p_{1m}^k(t_1,t_2)\right]^{\mathrm{T}}$.

Step 3: $\mathcal{N}_k$ is generated. $\boldsymbol{c}_{\mathrm{L}}^k$ is obtained by (9) and (10). If $\boldsymbol{c}_{\mathrm{L}}^k \geq \mathbf{c}_{\mathrm{L}}^{\min}$, $F_{\mathrm{b}} = F_{\mathrm{b}} + 1$, and *Step 7* is completed.

Step 4: The interaction mode $\boldsymbol{M}_{\mathrm{I}}^k$ is evaluated by (11). If $\boldsymbol{M}_{\mathrm{I}}^k = 0$, $\mathcal{N}_k' = \mathcal{N}_k$. If $\boldsymbol{M}_{\mathrm{I}}^k = 1$, $\mathcal{N}_k'$ is shown by the following equation:

$$\mathcal{N}_k' = \left\{l' \,\middle|\, l' \in \mathcal{N}_k, r_{l'} \leq p_{\mathrm{r}}, \boldsymbol{w}_{l'} > \boldsymbol{w}_k, \boldsymbol{w}_{l'} \leq \boldsymbol{w}_l, \forall l \in \mathcal{N}_k\right\} \tag{14}$$

where $r_{l'}$ is a uniformly distributed random number.

Step 5: The control input $\boldsymbol{u}_{ij}^k$ is calculated by (12). $\boldsymbol{P}^k$ is updated by (6) and (3).

Step 6: $\boldsymbol{y}^k(t)$ is updated by the following equation:

$$\boldsymbol{y}^k(t+1) = \sum_{l=1}^{N} a_k^l(t)\boldsymbol{y}^k(t) - \alpha\nabla f_k \tag{15}$$

where the alignment weight $a_k^l(t) = a_l^k(t)$ and $\sum_{l=1}^{N} a_k^l(t) = 1$. If $a_k^l(t) > 0$, then $\eta \leq a_k^l(t)$, where $0 < \eta < 1$ is the lower bound of the nonzero alignment weight.

Step 7: If $k = N$, *Step 8* is completed. Otherwise, $k = k+1$, and then *Step 3* is repeated.

Step 8: If $F_{\mathrm{b}} = N$, *Step 9* is completed.

Step 9: If $t_2 < T_{\max}^2$, $t_2 = t_2+1$, and then *Step 2* is repeated, where $T_{\max}^2$ is the maximum simulation time of the inner loop.

Step 10: $\boldsymbol{P}^k(t_1, T_{\max}^2)$ is output.

Step 11: If $t_1 < T_{\max}^1$, $t_1 = t_1 + 1$, $\boldsymbol{w}_k$ is updated by (7) and (8), and then *Step 2* is repeated, where $T_{\max}^1$ is the maximum simulation time of the outer loop.

The flowchart of the algorithm is illustrated in Fig. 4.

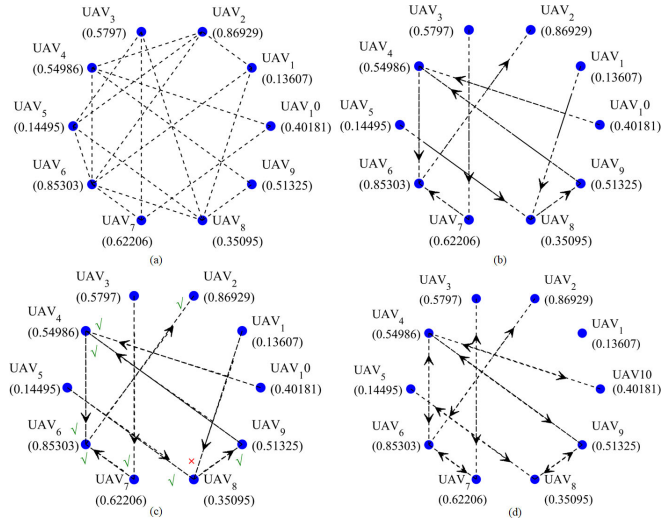

Fig. 3. Transformation process of interaction networks in the hierarchical interaction mode.

B. Convergence Proof

The following lemma is necessary for the convergence analysis of the proposed algorithm.

Lemma 1: ([28]): We assume that $\widehat{f_k}$ is a convex function of v_k, that t' denotes the termination time, that $\overline{\boldsymbol{y}}$ is the optimal solution, that $\alpha > 0$ is the step-size, and that the subgradient boundary $\widehat{L} \geq \left\|\nabla\widehat{f_k}\right\|$. Then

$$\lim_{t'\to\infty} \sum_{k=1}^{N} \widehat{f_k}\left(\frac{1}{t'}\sum_{t=0}^{t'-1}\boldsymbol{y}^k(t)\right) \leq \sum_{k=1}^{N} \widehat{f_k}(\overline{\boldsymbol{y}})$$

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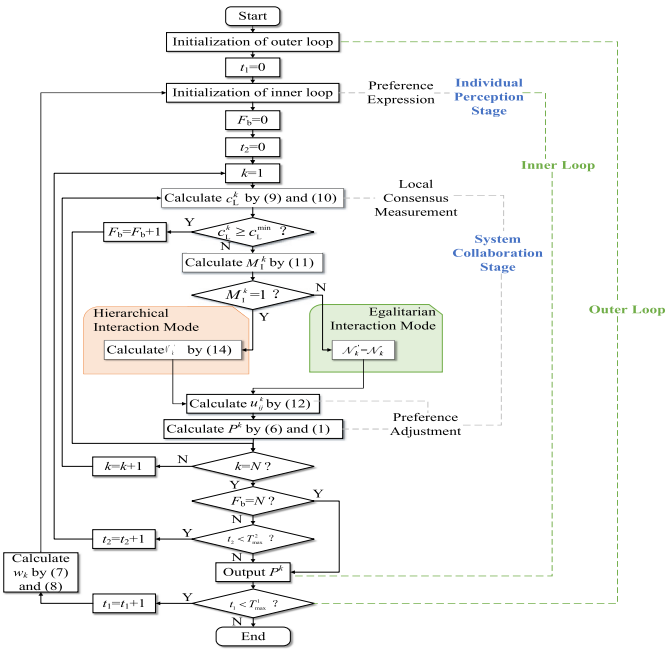

Fig. 4. Flow chart of the proposed algorithm.

$$+\alpha\widehat{L}^2\left(\frac{1}{2}+6N+\frac{6N^2}{1-\left(1-\eta^{(N-1)B}\right)^{\frac{1}{(N-1)B}}}\frac{1+\eta^{-(N-1)B}}{1-\eta^{(N-1)B}}\right) \tag{16}$$

when Condition 1 (in Section II) and $\max_{1\leq k\leq N}\left\|\boldsymbol{y}^k(0)\right\| \leq \alpha\widehat{L}$ are satisfied.

Theorem 1: : If $K_{\mathrm{F}} \geq \frac{1}{L}$ in $\boldsymbol{V}$ and the element p_{1j}^k in the fuzzy preference relation matrix is updated by (6), (11) and (12) under Condition 1, the error between the individual average estimated global objective function value and the global objective optimal value, as computed by $E_{\mathrm{r}} = f_k\left(\frac{1}{t'}\sum_{t_2=0}^{t'-1}\begin{bmatrix} p_{12}^k(t_1,t_2) \\ \vdots \\ p_{1M}^k(t_1,t_2)\end{bmatrix}\right) - f_k\left(\begin{bmatrix} p_{12}^*(t_1) \\ \vdots \\ p_{1M}^*(t_1)\end{bmatrix}\right)$, is bounded.

Proof: We assume that $\boldsymbol{X}^k = \boldsymbol{X}$ for all $k = 1,\ldots,N$. Equation (6) and (12) are transformed into the following equation:

$$\begin{aligned}
&\begin{bmatrix} p_{12}^k(t_1,t_2+1) \\ \vdots \\ p_{1M}^k(t_1,t_2+1)\end{bmatrix} \\
&= \begin{bmatrix} p_{1j}^k(t_1,t_2) + \sum_{l\in\mathcal{N}_k'}\min\{\frac{1}{n_d^k+1},\frac{1}{n_d^l+1}\} \\ \cdot[p_{12}^l(t_1,t_2) - p_{12}^k(t_1,t_2)] - K_{\mathrm{F}}\nabla_{p_{12}^k} f_k(p_{12}^k) \\ \vdots \\ p_{1M}^k(t_1,t_2) + \sum_{l\in\mathcal{N}_k'}\min\{\frac{1}{n_d^k+1},\frac{1}{n_d^l+1}\} \\ \cdot[p_{1M}^l(t_1,t_2) - p_{1M}^k(t_1,t_2)] - K_{\mathrm{F}}\nabla_{p_{1M}^k} f_k(p_{1M}^k)\end{bmatrix} \\
&= \sum_{l\in\mathcal{N}_k'}\min\{\frac{1}{n_d^k+1},\frac{1}{n_d^l+1}\}\begin{bmatrix} p_{12}^l(t_1,t_2) \\ \vdots \\ p_{1M}^l(t_1,t_2)\end{bmatrix} \\
&+ \left(1-\sum_{l\in\mathcal{N}_k'}\min\{\frac{1}{n_d^k+1},\frac{1}{n_d^l+1}\}\right)\begin{bmatrix} p_{12}^k(t_1,t_2) \\ \vdots \\ p_{1M}^k(t_1,t_2)\end{bmatrix} \\
&- K_{\mathrm{F}}\begin{bmatrix} \nabla_{p_{12}^k} f_k(p_{12}^k) \\ \vdots \\ \nabla_{p_{12}^k} f_k(p_{12}^k)\end{bmatrix}
\end{aligned} \tag{17}$$

Let $t = t_2$, $\widehat{f_k} = f_k$, and $\alpha = K_{\mathrm{F}}$; then (17) is transformed into the following equation:

$$\boldsymbol{y}^k(t+1) = \sum_{l=1}^{N} a_k^l(t)\boldsymbol{y}^k(t) - \alpha\nabla\widehat{f_k} \tag{18}$$

where a_k^l satisfies the equation:

$$a_k^l = \begin{cases} 1-\sum_{l\in\mathcal{N}_k'}\min\left\{\frac{1}{n_d^k+1},\frac{1}{n_d^l+1}\right\}, & l == k \\ \min\left\{\frac{1}{n_d^k+1},\frac{1}{n_d^l+1}\right\}, & l\in\mathcal{N}_k' \\ 0, & otherwise \end{cases} \tag{19}$$

From (19), $a_k^l(t) = a_l^k(t)$ and $\sum_{l=1}^{N} a_k^l(t) = 1$. The maximum degree of a UAV is $N-1$. From (19), $a_k^l \geq \frac{1}{N}$ and $\eta = \frac{1}{N}$.

From (5), $\left\|\nabla\widehat{f_k}\right\| \leq \sqrt{m-1}L$. From $p_{ij}^k \in [0,1]$, $\max_{1\leq k\leq N}\left\|\boldsymbol{y}^k(0)\right\| \leq \sqrt{m-1}$. From $K_{\mathrm{F}} \geq \frac{1}{L}$, $\max_{1\leq k\leq N}\left\|\boldsymbol{y}^k(0)\right\| \leq \alpha\widehat{L}$, where $\alpha = K_{\mathrm{F}} \geq \frac{1}{L}$ and $\widehat{L} = \sqrt{m-1}L$. According to Lemma 1, (16) is satisfied:

$$\begin{aligned}
&\lim_{t'\to\infty}\sum_{k=1}^{N} f_k\left(\frac{1}{t'}\sum_{t_2=0}^{t'-1}\begin{bmatrix} p_{12}^k(t_1,t_2) \\ \vdots \\ p_{1M}^k(t_1,t_2)\end{bmatrix}\right) \\
&\leq \sum_{k=1}^{N} f_k\left(\begin{bmatrix} p_{12}^*(t_1) \\ \vdots \\ p_{1M}^*(t_1)\end{bmatrix}\right) + E_{\mathrm{r}} \\
&= \sum_{k=1}^{N} f_k\left(\begin{bmatrix} p_{12}^*(t_1) \\ \vdots \\ p_{1M}^*(t_1)\end{bmatrix}\right) \\
&\quad + (M-1)K_{\mathrm{F}}L^2\left(\frac{1}{2}+6N+6N^2A\right)
\end{aligned} \tag{20}$$

where the parameter $A = \frac{1}{1-\left(1-\eta^{(N-1)B}\right)^{\frac{1}{(N-1)B}}}\cdot\frac{1+\eta^{-(N-1)B}}{1-\eta^{(N-1)B}}$. Thus, $E_{\mathrm{r}} = (M-1)K_{\mathrm{F}}L^2(\frac{1}{2}+6N+6N^2A)$ is bounded.

C. Error Impact Analysis

In order to enhance the applicability of the proposed approach, an analysis of the impact of errors is conducted, and several potential improvement methods are summarized.

Given (2), M, L, and N are fixed. According to (20), the smaller K_{F} and A are, the smaller E_{r} is, where A is a function related to the lower bound of the nonzero alignment weight η and intercommunication interval B. The minimum value of

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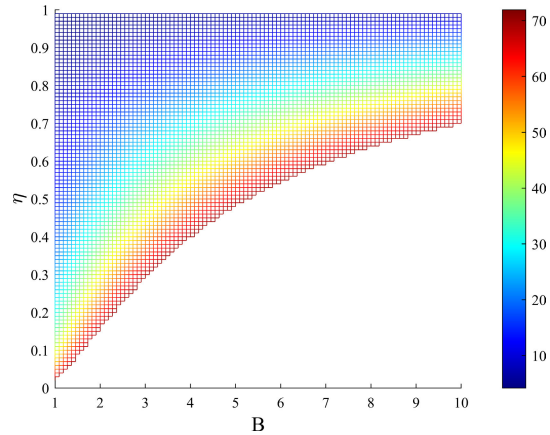

Fig. 5. Values of $\log_{10}(A)$.

K_F is $\frac{1}{L}$. Thus, a greater η leads to a smaller A and E_r. A greater B leads to a greater A and E_r. According to (19), the denser the network is, the greater the node's degree, the smaller the alignment weight, and the smaller the minimum value. A denser network also means a smaller reconnection interval between two nodes. There is a positive correlation between η and B, but the effect of η on E_r is opposite to that of B on E_r to some extent. The values of A corresponding to different η and B are shown in Fig. 5, where $N = 10$. As shown, the larger the value of η is, the less sensitive the value of A to the change in B. This finding suggests that a smaller A could be obtained by increasing η at the expense of B, as achieved by (14) in the hierarchical interaction model.

From (14), the actual network is determined by the available neighbor set $\mathcal{N}_k$, the acceptance probability p_r, and consensus weight $\boldsymbol{w}$. $\mathcal{N}_k$ is determined by the physical environment. A smaller p_r and a $\boldsymbol{w}$ with a great difference in elements may mean a larger η and B. A $\boldsymbol{w}$ that reflects the individuals' initial quality will slightly retain the effective interaction between individuals and high-level individuals to a certain extent and reduce the interference from low-level individuals. In addition, the coincidence of $\boldsymbol{w}_k$ and $\boldsymbol{I}_k$ could also reduce the difference between $f_k(\hat{p}^k_{1j}, p^k_{1j})$ and the real individual objective optimization function $f_k(p^*_{1j}, p^k_{1j})$.

In summary, E_r can be reduced in the following ways.

1) Use of the objective optimization control gain $K_F = \frac{1}{L}$.
2) Selection of an appropriate acceptance probability p_r to maximize the nonzero alignment weight η when the increase in B is small or zero.
3) Selection of appropriate PID parameters or other outer loop control methods to make $\boldsymbol{w}_k = \boldsymbol{I}_k$.

V. Simulation Results

To test the feasibility of our proposed method, numerical results are provided. Specifically, we evaluate the performance of the algorithm under different conditions. Comparative simulations based on the DGD algorithm [28] are conducted to further verify the superiority of the algorithm. The effectiveness of the outer loop design is also validated. Moreover, we provide an analysis of the impact of errors and discussed potential improvements. The simulation is executed on Lenovo X1 Carbon with an Intel Core i5 CPU at 2.30 GHz, and an 8G MEM.

A. Parameter Settings

We assume that $\boldsymbol{X}^k = \boldsymbol{X}$ is satisfied for all UAVs and the graph is produced by the Erdos-Renyi model [47] with a connectivity probability of 0.7. The simulation is conducted based on the specific steps of the proposed coordination algorithm. The values of the parameters are set as $N = 10$, $M = 10$, $K_F = 0.1$, $\boldsymbol{I}_k \in (0, 1)$, $\mathbf{c}_L^{\min} = 1$, $\mathbf{c}_L^{S} = 0.998$, $T^1_{\max} = 10$, $T^1_{\max} = 100$, $K_P = 1$, $K_I = 0$, $K_D = 0$, $\beta = 1$, $p_r = 0.2$.

Then, we construct the real fuzzy preference relation matrix $\boldsymbol{P}^*$, regarded as a complete information matrix provided by real situation, by the following equation:

$$p^*_{ij}(t_1) = \begin{cases} [1 + \pi_i(t_1) - \pi_j(t_1)]/2, & if\ i = 1, j = 2, \ldots, m \\ 0.5 + p^*_{1j}(t_1) - p^*_{1i}(t_1), & otherwise \end{cases} \tag{21}$$

where $\pi_i = b_i / \sum_{i=1}^{M} b_i$ is the normalized profit of x_i, and b_i is set to obey the normal distribution with mean $T^2_{\max}(i-1)/M$ and variance $T^2_{\max}/M$.

To accurately assess the performances of the algorithm, three evaluation criteria are defined, and their specific computation methods and effects are given through the following remarks.

Remark 1: The global consensus parameter c_G reflects the level of relative consensus, which is computed by

$$c_G = \frac{\left\| \sum_{k=1}^{N} \left([p^k_{12} p^k_{13} \cdots\ p^k_{1m}] / \left\| [p^k_{12} p^k_{13}\ \cdots p^k_{1m}] \right\| \right) \right\|}{N} \tag{22}$$

where $\|\|$ denotes the norm of the vector. The result of summing the normalized vectors reflects the degree of correlation of each vector. Thus, the higher the value of c_G (approaching 1) is, the higher the level of relative consensus.

Remark 2: The swarm objective function F reflects the level of absolute consensus, which is computed by

$$F = \sum_{k=1}^{N} f_k(\hat{p}^k_{1j}, p^k_{1j}). \tag{23}$$

where the lower the value of F is, the higher the level of group optimization. *Remark 3*: Another criterion used to assess the level of absolute consensus is the optimization of the real swarm objectives, denoted as F^*, which is computed by

$$F^* = \sum_{k=1}^{N} f_k(p^*_{1j}, p^k_{1j}). \tag{24}$$

where the lower the value of F^* is, the higher the level of absolute optimization.

B. Performance Under Ideal Conditions

When testing the inner loop algorithm, $T^1_{\max}$ is set to 1 and $\boldsymbol{w}_k = \boldsymbol{I}_k / \sum_{k=1}^{N} \boldsymbol{I}_k$. The initial fuzzy preference relation matrices $\mathbf{P}^k(0, 0)$ are given in Fig. 6; they have a significant difference. The initially generated matrices conform to the properties.

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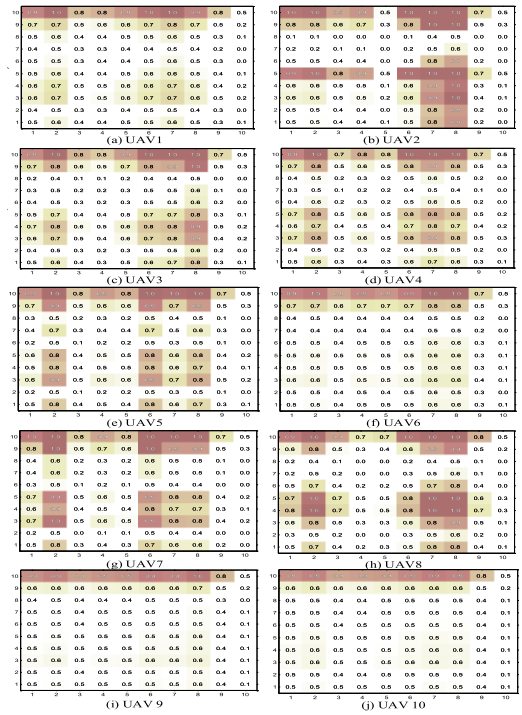

Fig. 6. Initial fuzzy preference relation matrix $\boldsymbol{P}^k(0, 0)$.

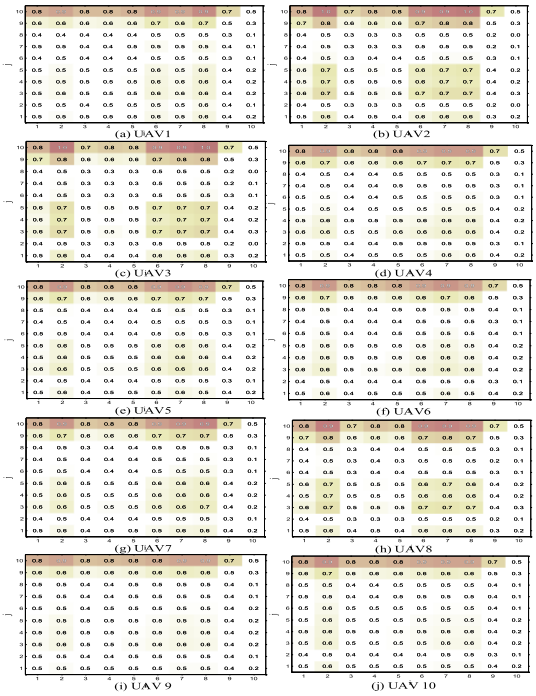

Fig. 7. Final fuzzy preference relation matrix $\boldsymbol{P}^k(0, T^2_{\max})$.

The final fuzzy preference relation matrices $\boldsymbol{P}^k(0, T^2_{\max})$ are shown in Fig. 7. Notably, the final matrices are similar. The swarm system reaches a relative consensus on SA.

Fig. 8 describes the curves of f_k under ideal conditions, where f_k gradually converges to a similar value, that is, individual SA in the system is basically at the same level.

C. Performance Under Nonideal Conditions

To further test the feasibility of the inner loop algorithm under nonideal conditions, the time delay in the neighbor

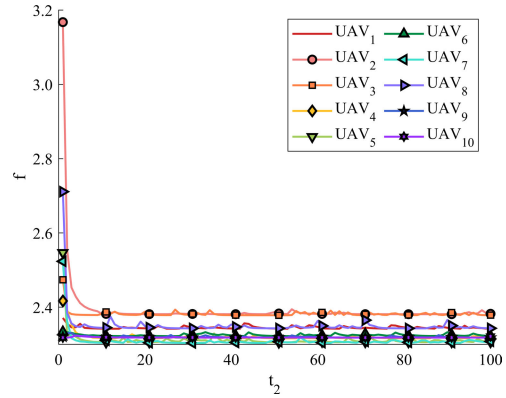

Fig. 8. Curves of f_k under ideal conditions.

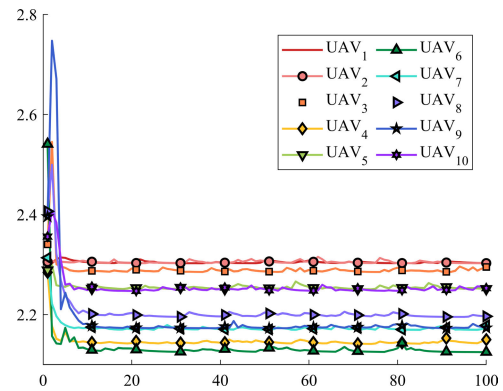

Fig. 9. Curves of f_k under nonideal conditions.

interaction process is considered:

$$\boldsymbol{u}^k_{1j} = \sum_{l \in \mathcal{N}'_k \& x_j \in \boldsymbol{X}^l} \min\left\{\frac{1}{n^k_d + 1}, \frac{1}{n^l_d + 1}\right\} \cdot \left(p^l_{1j}(t_1, t_2 - t_{\mathrm{d}}) - p^k_{1j}\right) - K_{\mathrm{F}} \nabla_{p^k_{1j}} f_k(p^k_{1j}). \quad (25)$$

where t_{d} is the time delay of neighbor l. Here, t_{d} is set to 1 with a probability of 0.1; otherwise, it is 0. Other parameters are the same as the above ideal conditions.

Fig. 9 describes the curves of f_k with interaction delay. Despite a large fluctuation in the initial stage, f_k is stable at a similar value. Thus, the algorithm is still feasible, although the introduction of a time delay results in the loss of SA consensus.

D. Comparison With the DGD Algorithm

A comparative simulation is conducted to evaluate and validate the performance of the proposed algorithm, demonstrating its superiority over DGD. For fairness, the step size of DGD is set to K_{F}, and the other parameters are consistent with the above ideal condition. Fig. 10(a) and Fig. 10(b) describe the comparative change in c_{G} and F.

Notably, the proposed algorithm is superior in terms of the convergence value of c_{G}. The convergence state is smaller than DGD in both amplitude and frequency. In addition, the final value of c_{G} is approximately 0.0025 greater than that of DGD, indicating a higher quality relative consensus of SA. As shown in Fig. 10(b), the curve of the proposed algorithm is

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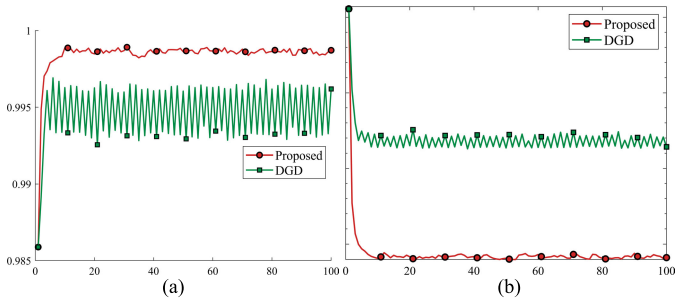

Fig. 10. Comparative results of the inner loop: (a) c_G and (b) F.

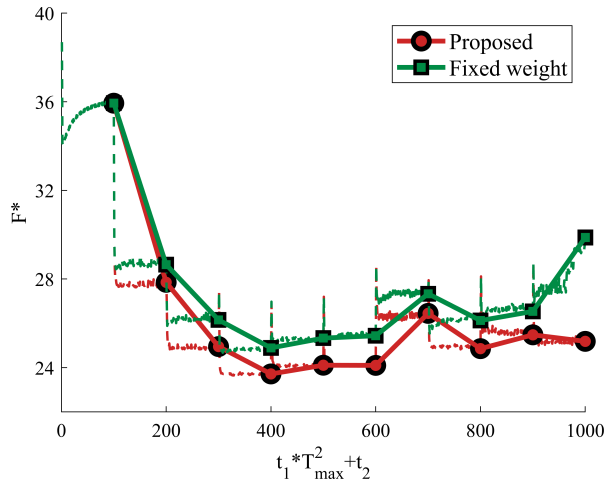

Fig. 11. Comparison of F^*.

always below that of DGD. The proposed algorithm's function F is approximately 0.8 less than that of DGD, indicating that the result of the proposed algorithm is more suitable for the real environment. It is worth noting that the slight fluctuations observed in the convergence curves of both c_G and F can be attributed to the influence of errors, which can be mitigated by implementing the approaches outlined in the preceding section.

E. Effectiveness of the Outer Loop Design

To verify the necessity of the outer loop, comparative simulations based on a fixed $\boldsymbol{w}_k$ are conducted. For fairness, $\boldsymbol{w}_k$ is equal to the initial weight of the proposed algorithm, and the other parameters remain unchanged. Fig. 11 depicts the curves of F^* based on the proposed algorithm and the algorithm with fixed consensus weights.

Despite the consistent presence of marginal fluctuations in the curve of the proposed method, influenced by errors as illustrated in Fig. 11, it is noteworthy that the function value remains below that of the algorithm employing fixed consensus weights. This finding indicates that $\boldsymbol{P}^k$ of the proposed algorithm is more similar to $\boldsymbol{P}^*$ than that of the algorithm with fixed consensus weights.

F. Discussion

While the proposed algorithm demonstrates superior performance in achieving DSA consensus, it is crucial to acknowledge and address the potential difficulties and challenges that may arise in real-world applications.

1) The impact of uncertainties. Uncertainties in the application scenarios primarily manifest in two aspects: the initial level of individual cognition and the communication network. The former depends on the individual's sensor performance and the processing capability of sensor data, whereas the latter relies on communication and networks routing techniques.
2) Different modes in the algorithm. As mentioned, the proposed algorithm includes two interaction modes. In the egalitarian mode, individuals interact indiscriminately with neighbors within the communication range, which may lead to communication congestion and potential blockage in densely populated groups. On the other hand, the hierarchical mode, while ensuring communication with higher-ranking individuals, compromises the robustness of group connectivity. Thus, it is essential for the network to satisfy the requirements of Condition 1 in order for the algorithm to converge effectively.

VI. Conclusion

This paper addresses the distributed SA consensus problem for UAV swarm systems. We establish the system model by introducing the fuzzy preference relation theory and a time-varying undirected graph. By formulating a distributed optimization-based problem, it becomes possible to achieve both relative and absolute consensus. The proposed dual-loop decision-making framework incorporates a feedback mechanism, ensuring robustness against disturbances during the consensus process. Coupled with the dual-mode mechanism in the novel coordination algorithm, optimization relies on historical information feedback rather than individual iterations, thereby alleviating communication burdens among heterogeneous UAVs. The feasibility and effectiveness of our proposed methods are validated through theoretical analysis and simulation results.

In future work, UAV dynamics could be considered. Further clarification of the actual attributes of the target in real-world scenarios, including its geographical locations, dynamic properties, and levels of importance, would be advantageous. More current existing algorithms will be taken into account in the comparative simulations.

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